

EMMA LINSOMB

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BIO

Undergraduate researcher at Cornell University with experience in numerical relativity, computational physics, observational astronomy, and observational cosmology instrumentation. Seeking PhD programs in astrophysics, with particular interest in gravitational wave astrophysics or astroparticle physics.

EDUCATION

Cornell University, Ithaca, NY

B.A., Physics, 2023 to 2027 (expc.)

Thesis Title: Examining Tidal Deformability of Neutron Stars by Simulating Inspiral Degeneracy of Differing Internal Structures

Select Coursework: Introduction to Computing: Design and Development in Python • Waves and Thermodynamics • Analytical Mechanics • Advanced Electricity and Magnetism • Intermediate Quantum Mechanics • Introduction to Particle Physics • Multiwavelength Astronomical Techniques • Multivariable Calculus • Differential Equations • Linear Algebra • Applied Complex Analysis • Partial Differential Equations • Science in American Politics

GPA: 3.91/4.3

RESEARCH EXPERIENCE

Professor Nils Deppe Computational Relativity Research Group

Cornell University

Research Assistant

Aug 2025 - Present

- Implemented modified Tolman–Oppenheimer–Volkoff Equations in Python to numerically solve theoretical neutron stars with different equations of state (EOS).
- Performed rigorous error analysis to determine the validity of simulated solutions.
- Wrote C++ code that contributed to the SpECTRE project and improved the accuracy of neutron star solutions for polytropic EOS, using a rigorous expansion near the stellar center.
- Investigated the waveforms of neutron star binaries with degenerate tidal deformabilities, but different internal EOS.

Dr. Amanda Bauer Observational Astronomy Research Group

Yerkes Observatory

Research Assistant

May 2025 - Aug 2025

- Examined the growth of the Dumbbell Planetary Nebula (M27) over a period of 119 years.
- Operated the Yerkes 41" Reflector for data collection and the Yerkes 40" Great Refractor for optical observations.
- Handled, digitized, and inspected historical glass plate observations from the Yerkes 180,000 glass plate collection.
- Utilized astropy, photutils, and publicly available astronomy programs such as SAO DS9 and AstroImageJ to compare historical glass plate observations to modern-day digital observations.
- Presented final findings in a 15-minute presentation to the Yerkes board of directors, and wrote a final paper documenting methods and results.

Professor Abigail Crites Observational Cosmology Research Group

Cornell University

Research Assistant

Jan 2024 - May 2025

- Examined [CII] emission spectra from the early universe.

- Implemented code for the calibration of the emission spectra detectors.
- Installed emission spectra detectors and tested wiring throughout the instrument.
- Operated heavy machinery to assemble and disassemble the instrument.
- Collaborated with a team of 4 other undergrads on mechanical issues in the instrument.
- Presented final results to an audience of my peers during the Cornell Nexus Scholars summer program.

AWARDS AND FELLOWSHIPS

Cornell Arts & Sciences Nexus Scholar (\$7,000) 2024

Fellowship funding a 9-week, full-time summer research experience at Cornell

SPS Association for Women in Science Kirsten R. Lorentzen Award (\$2,500) 2026

This award is given annually to one exceptionally well-rounded student who excels in her studies as well as outdoor activities, service, sports, music, or other non-academic pursuits or who has overcome significant obstacles.

PRESENTATIONS

247th American Astronomical Society Meeting 01/2026

Poster: *Examining the Growth of the Dumbbell Nebula, M27, with 119 Years of Observations*

Cornell Undergraduate Physics Poster Session 12/2025

Poster: *Examining the Growth of the Dumbbell Nebula, M27, with 119 Years of Observations*

Yerkes Board of Directors Colloquium 08/2025

Talk: *Measuring Growth of the Dumbbell Nebula*

Cornell Nexus Scholars Capstone 07/2024

Talk: *Optimization and Lab Characterization of the TIME Instrument*

MENTORSHIP

Cornell Physics Department Learning Assistant Program **Cornell University**

Learning Assistant 01/2026 - Present

- Attended weekly classes focused on developing teaching skills.
- Worked with the course team during weekly meetings to address the content of the course and the performance of the students.
- Spent ~ 2 hours per week running office hours and assisting students in developing their conceptual understanding of the course content.

Society of Physics Students Peer Mentorship Program **Cornell University**

Peer Mentor 08/2024 - Present

- Mentored 7 students total: 3 in the 2024-2025 academic year, 4 in the 2025-2026 academic year.
- Met with freshmen mentees once a month and guided them through the process of adjusting to the physics major and college life in general.

SCIENCE LEADERSHIP AND SERVICE

Cornell Society of Physics Students

Cornell University

President

12/2025 - Present

- Managed about a dozen events per semester, including speaker events, career panels, socials, mentorship programs, local outreach, the Cornell Physics Wiki, and an annual poster session.
- Served on the CEDI committee, and continue to facilitate community-building initiatives in the Cornell Physics Department (more information below).
- Organized the initiatives of the organization, promoting professional development and community.
- Planned the third-annual AEPxPhysics Formal in collaboration with the Applied Engineering Physics Society and the Physics Graduate Society, taking on the primary leadership role for the planning committee.

Vice President

12/2024 - 12/2025

- Organized and oversaw ~ 10 physics events per semester, collaborating with the president and executive board.
- Led our undergraduate peer mentorship program of ~ 100 participants and our undergraduate-graduate mentorship program with ~ 12 participants.
- Served on the CEDI committee and completed the “A Physicist Is” project for the second cohort (more information below).
- Planned and hosted the second-annual AEPxPhysics Formal, working with the Applied Engineering Physics Society.
- Wrote an article for the SPS Observer, featured in the Winter 2025 issue.

Women in Physics Chair

02/2024 - 12/2024

- Planned and hosted ~ 10 physics events per year: including professional development workshops, speakers, socials, etc.
- Served as a student-faculty liaison on the Cornell Physics Committee for Climate, Equity, Diversity, and Inclusion (CEDI).
- Developed “A Physicist Is” under Prof. Holmes, a project to showcase the diversity of careers and identities in physics through informational graphics posted around the Physical Sciences Building.

OTHER LEADERSHIP AND SERVICE

Hearsay A Cappella

Cornell University

Alumni Chair

5/2026 - Present

- Kept in constant contact with Hearsay alumni to facilitate travel to future concerts, get donations, and support community efforts.

Choreography Chair

5/2025 - Present

- Designed choreography for all songs in the set-list, and led dancing-focused rehearsals for 15 members.
- Worked closely with the music director and president to schedule and plan extra rehearsals in the weeks leading up to the concert.

- Held weekly office hours for members needing help with their choreography.
- Filmed practice runs of the show and wrote individualized notes for every member based on their performance.

Assistant Music Director

5/2025 - Present

- Arranged multiple pieces for performance and taught newer members how to read and write music.
- Led singing-focused rehearsals for 15 members when the main music director was absent, and assisted the main music director with leading warm-ups and giving performance feedback.
- Coordinated group auditions with the group's president, business manager, and music director.
- Held weekly office hours for members needing help learning their music or preparing for their solos.
- Taught members how to beatbox.

Internal Social Chair

5/2024 - 12/2024

- Planned and hosted social events for members throughout the semester, including new member celebration, senior sendoff, and mid-semester date night.
- Organized the big-little program.

SKILLS

Programming: Python (NumPy, SciPy, Matplotlib, pandas), C++, Git, Bash, L^AT_EX, Windows, MacOS

Scientific Computing: Numerical Methods, Error Analysis, Scientific Computing, Computational Physics, Data Analysis

Astrophysics Software: AstroPy, MaximDL, AstroImageJ, SAODS9, Siril

Experimental & Observational: Telescope Operation, Detector Calibration, Scientific Instrumentation, Archival Astronomical Data, Image Reduction, Instrument Assembly

Research: Scientific Writing, Literature Review, Technical Presentations, Collaborative Research

Language: Native Fluency in English, B2 Fluency in French

Management: Teaching, Mentorship, Budget Management, Leadership, Large Event Planning, Delegation

AFFILIATIONS

Member, American Astronomical Society

Member, American Physical Society

Member, Society of Physics Students